ELSEVIER

Evolution and Human Behavior 31 (2010) 39–47

Evolution and Human Behavior

Sexual selection under parental choice in agropastoral societies

Menelaos Apostolou

Department of Psychology, University of Warwick, CV4 7AL Coventry, UK

Initial receipt 7 January 2009; final revision received 22 June 2009

Abstract

Evidence from the anthropological record indicates that in most human societies, parents control the mating access to their offspring. Based on these data, a model of sexual selection has been recently proposed, whereby along with female and male choice, parental choice constitutes a significant sexual selection force in our species. This model was found to provide a good account for the mating patterns which are typical of foraging societies. By employing data from the Standard Cross Cultural Sample, the present study aims at examining whether this model can also account for the mating patterns typical of agricultural and pastoral societies. In addition, comparisons between different society types are made and two model-derived hypotheses are tested. First, it is hypothesised that parents have more control over their offspring's mate choices in non-foraging societies. Second, it is hypothesised that male parents exert greater decision making power in agropastoral societies than in hunting and gathering ones. Both hypotheses are supported by the results presented here. The evolutionary implications of these findings are also explored.
© 2010 Elsevier Inc. All rights reserved.

Keywords: Sexual selection; Parental choice; Mating behaviour; Mating patterns in Agricultural Societies; Mating patterns in Pastoral Societies

1. Introduction

In the majority of human societies parents have a substantial role in determining who is going to have mating access to their offspring. As the anthropological record indicates, the offspring are constrained from exercising mate choice freely and the selection of spouses is frequently made by their parents (Apostolou, 2007b; Broude & Green, 1983; Minturn, Grosse, & Haider, 1969; Stephens, 1963; Westermarck, 1925; Whyte, 1978). Based on this evidence, Apostolou (2007b) proposed a model whereby, along with female and male choice, parental choice constitutes a significant sexual selection force in our species.

This model was found to be consistent with the patterns of mating typical of modern foraging societies. However, given that most preindustrial societies base their subsistence on agriculture and animal husbandry (Murdock, 1981), understanding the workings of sexual selection today requires that the analysis is not restricted solely to foraging societies. Accordingly, this article aims at identifying the typical patterns of mating in contemporary agricultural and pastoral societies, and examine whether these patterns are consistent with the model of parental choice.

Furthermore, anthropological evidence from contemporary societies provides valuable insights on how sexual selection had been working in ancestral human societies (Ember, 1978; Lee & DeVore, 1968). As modern agropastoral societies resemble ancestral ones, which appeared 10,000 years ago, the identification of the typical patterns of mating in the former enables us to infer the typical patterns of mating in the latter. And even though 10,000 years is a rather short time period in the evolutionary timescale, this later stage of human evolution is significant because, if certain conditions are met, substantial evolutionary change can take place in only a few generations.

In particular, two conditions can trigger rapid evolutionary change; namely the strength of the selection pressure and the genetic variability available in the population (Fisher, 1958). These conditions were satisfied at the onset of the Holocene and the transition from a social organisation based on hunting and gathering to a social organisation based on agriculture and animal husbandry. First, demographic growth accelerated substantially commencing 10,000 years ago in the Middle East and Asia spreading into Europe and Australasia in the next 6000 years (Price, 2000). Conse-

E-mail address: m.apostolou@gmail.com.

1090-5138/$ – see front matter © 2010 Elsevier Inc. All rights reserved.
doi:10.1016/j.evolhumbehav.2009.06.010

quently, this population growth provided the necessary genetic variability required for rapid evolution to take place.

Second, the agropastoral revolution had a subsequent dramatic change in the environment of human evolution, as humans started leading a sedentary life instead of a nomadic one and their social environment was no longer a small band of people, but villages and cities of hundreds or thousands. It is no surprise that Hawks, Wang, Cochran, Harpending, and Moyzis (2007) found evidence of rapid acceleration of genetic evolution in our species and attributed this to changes in human cultures and the rapid demographic growth over the last few thousand years (see also Armelagos & Harper, 2005).

Overall, if shifting economic structures caused changes in the patterns of mating, the right circumstances were there for directional selection to take place. Accordingly, this study further aims at examining whether the change in the mode of subsistence had an effect on human mating by comparing the patterns of mating in agropastoral societies with the respective patterns of matting in foraging societies in order to identify recurring differences between the two.

2. The model of parental choice

Parents and offspring are not genetically identical and as a consequence they do not share identical interests with respect to mating (Trivers, 1974). Specific traits in a mating candidate give asymmetrical benefits to each party resulting into conflict between the two. Research has revealed disagreement over genetic quality, family background, religiousness and personality (Apostolou, 2007a, 2008b, 2008c; Buunk, Park, & Dubbs, 2008). Therefore, the choice of a spouse that maximises the inclusive fitness of the offspring does not necessarily maximise the inclusive fitness of their parents. Thus, asymmetric interests with respect to mating induce parents to control their offspring's mating decisions and select in-laws who maximise their own inclusive fitness.

Parental control over mating is made possible mainly by two factors: the extensive period that the offspring are dependent upon their parents' investment for survival and reproduction, and the fact that parents and their kin are physically stronger than their children (Apostolou, 2007b). Thus, parents can withhold their resources and/or apply physical punishment so as to take control of their offspring's mating decisions. As parents are able to influence their children's mate choices, parental choice becomes a significant sexual selection mechanism: traits that make an individual appealing to parents are likely to be selected and increase in frequency in the population.

Furthermore, parental control over mating is asymmetrical, with parents exercising more control over their daughters' mating decisions (Apostolou, 2007b; Perilloux, Fleischman, & Buss, 2008). There are two main reasons that account for this asymmetry: first, the female, by investing more in her offspring, becomes the scarce reproductive resource to which males are seeking access (Trivers, 1972). Accordingly, by controlling the mating behaviour of their daughters, parents effectively gain control over mate choice. Second, relaxation of parental control over the female offspring may potentially have severe consequences for the parents: a sexual adventure can commit her to a long period of investment (i.e. pregnancy) in the offspring of a man whom her parents do not necessarily approve. It should also be added that women are physically weaker than men, which makes them more susceptible to parental manipulation.

Parental choice is also asymmetrical in favour of male parents. By means of greater physical strength, exclusive use of weaponry and control of political institutions (Flinn & Low, 1986), male parents have more influence over the offspring's mate choices than their female spouses (Apostolou, 2007b; see also Smuts, 1995). Moreover, control over weapons and political institutions enables males to retain control over their children, and especially the female ones, even when they get old and can no longer impose their will by means of their physical strength.

Parental control over the offspring's mate choices has its limits, as the offspring have space to exercise their own choice. Parents cannot always be present guarding and monitoring the actions of their children. Furthermore, parental control wanes as the offspring grow-up and become physically stronger, while their parents grow older and become physically weaker. Also, as they age, the offspring become more experienced in subsistence activities, and consequently, less dependent upon their parents for survival. Additionally, the offspring can exercise mate choice in later marriages where their parents and close kin may be absent due to death, or less able to impose their will due to old age (Apostolou, 2007b). Finally, the offspring are not simply pawns in the hands of their parents as they can evolve adaptations to psychologically manipulate their parents towards their own ends (Trivers, 1974).

2.1. Parental choice in agricultural and pastoral societies

In the model of parental choice, when and where they can, parents exert control over their offspring's mating decisions. This strategy, and consequently the strength of parental choice as a significant sexual selection force, is contingent upon the ecological niche a population occupies. More specifically, in technology based post-industrial societies, such as the United Kingdom or the United States, wherein extensive schooling and training is required, marriage is postponed until the conclusion of one's education. Therefore, individuals tend to get married later in adulthood when they are no longer financially dependent on their parents. As a result, free mate choice is the typical pattern of mating in these societies.

On the other hand, in preindustrial agropastoral societies the offspring are heavily dependent upon parental investment, and parents are able to use physical force on their

offspring. In particular, food production is based on land and domesticated animals, which are privately owned and difficult to obtain. Also, because adult skills sufficient for economic independence can be acquired more rapidly than is true in most industrialized nations, education needs are not an impediment to betrothing. Finally, social protection systems like a welfare state or a national health system are hardly ever found in pre-industrial societies, where individuals rely on their close kin for support and protection. On these grounds, it is predicted that in these societies parents are able and do exercise substantial control over their offspring's mating decisions.

Moreover, in the model of parental choice the strength of parental control over mating is a function of the level of parental investment. Agricultural and pastoral societies have more material resources than foraging societies; subsequently, parents in these societies control more parental investment (land, cattle, etc.) which they can divert to their offspring. For instance, parental wealth can be used to finance a desirable marriage (Apostolou, 2008a) or be passed directly to the offspring in the form of inheritance. As these resources are quite substantial, the offspring are expected to be more willing to comply with their parents' wishes so as to avoid forfeiting parental investment.

In addition, since in agropastoral societies families control more wealth, parents have more to lose from allowing their offspring to exercise choice freely and more to gain from arranging their marriage. Accordingly, as the opportunity cost of free choice is higher in agropastoral societies, so must be parents' eagerness to control their offspring's mating decisions. Overall, because parents direct more resources in agropastoral than in foraging societies, it is predicted that they will exercise more influence over their offspring's mating behaviour in the former than in the latter.

Finally, foraging societies, in contrast to other preindustrial societies, are more egalitarian and females enjoy higher social status (Endicott, 1999; Whyte, 1978). Also, in agropastoral societies, resources are predominantly controlled by men despite the fact that women have a significant role in production (Lerner, 1986; Whyte, 1978). Higher social status combined with more control over wealth enable men to be more influential over their offspring's mating decisions. Thus, it is further predicted that male parents in agropastoral societies have more decision making power over their offspring's mating behaviour than male parents in foraging societies.

3. Method

For the purposes of this study the Standard Cross-Cultural Sample (SCCS), which includes 186 pre-industrial societies, was employed. Data were taken from the updated electronic version of SCCS from the World Cultures CDROM (World Cultures, 2006). Some of the variables had to be modified, as explained below, in order to test the specific hypotheses of this study.

In particular, the marriage arrangements variable was derived by collapsing a previous variable with more levels

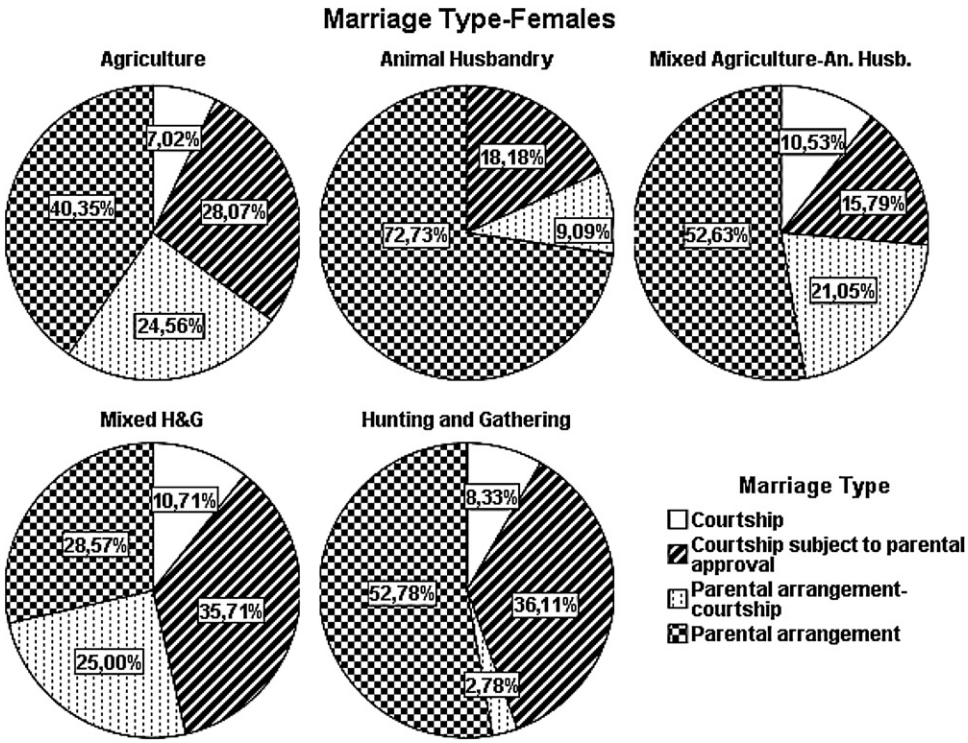


Fig. 1. The figures above depict the various types of marriage for females across different modes of subsistence. We can see that marriage based on courtship is always in the minority as opposed to parental arrangement marriage which is the most frequent type.

into the four levels of interest here: parental arrangement, courtship subject to parental approval, courtship, and parental arrangement-courtship. In the parental arrangement type of marriage, parents choose the marriage partners of their offspring; in the courtship subject to parental approval type, individuals find their own spouses but their choice is subject to their parents' approval; in the courtship type of marriage, individuals choose their own spouses with little or no influence from their parents, and finally in the parental arrangement-courtship type, both types of marriage are practiced in a given society.

Moreover, a new variable was constructed from previous codes, which reflects the level of punishment when a female is discovered to commit adultery. This variable has three categories: no punishment or light punishment, moderate punishment which includes beating and divorce, and severe punishment which includes severe beating that may result into death.

Based on existing codes, the societies in the sample were classified into five categories: *Hunting and gathering*: societies that base less than 10% of their subsistence on agriculture and less than 10% on animal husbandry; *Animal husbandry*: societies that base more than 50% of their subsistence on animal husbandry; *Agriculture*: societies that base more than 50% of their subsistence on agriculture; *Mixed agriculture-animal husbandry*: societies that base more than 10% of their subsistence on animal husbandry and/or agriculture with contributions of less than 10% from hunting, gathering and fishing respectively; *Mixed hunting and gathering*: societies that base more than 10% of their subsistence on animal husbandry and/or agriculture with contributions of more than 10% from hunting and/or gathering and/or fishing.

4. Results

4.1. Control of marital decisions

Similarly to hunting and gathering societies, parental arrangement is the typical mode of marriage in agricultural and pastoral societies. More specifically, with respect to women's marriage, it can be seen from Fig. 1 that parental arrangement is the most frequent type of marriage in agricultural, animal husbandry and mixed agriculture-animal husbandry societies. This is also the case in hunting and gathering societies, with the exception of mixed hunting and gathering ones where parental arrangement comes second in frequency, after the courtship subject to parental approval type of marriage. The latter is the second most frequent type of marriage in agricultural and pastoral societies. Furthermore, parental arrangement-courtship is the third most frequent marriage type in the sample. Finally, courtship is the least frequent type of marriage across all society types.

In Fig. 2 we can see somewhat different results with respect to the marriage of men. Parental arrangement is again the most frequent type of marriage across the various society

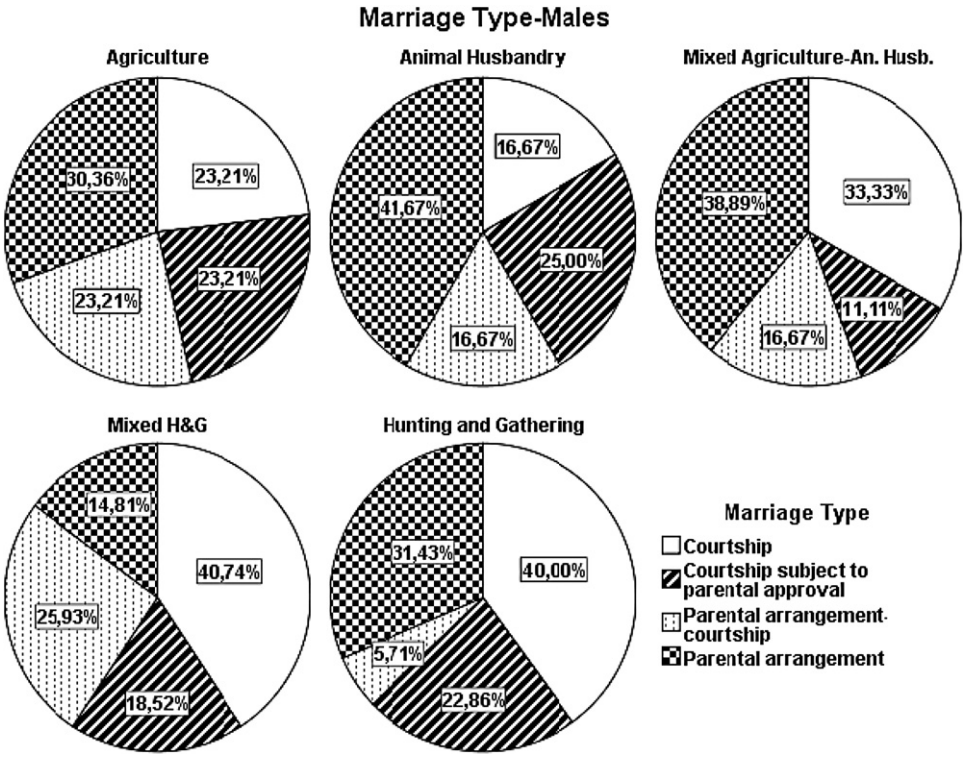


Fig. 2. The figures above depict the various types of marriage for males across different modes of subsistence. We can see that although parental arrangement is common across different society types, marriage based on courtship is also very common, especially in hunting and gathering societies.

types, with the only exception of the mixed hunting and gathering societies. However, in comparison to the marriage of females, the courtship type of marriage for males is higher in frequency across all groups of societies. Additionally, courtship marriage for men is less frequent, and parental arrangement more frequent in agricultural and pastoral societies than it is in hunting and gathering ones.

To test whether the marriage type depends on the sex of the spouse, multinomial logistic regression was employed. Initially the type of marriage was regressed on the sex of the spouse. The results indicate that these two variables are not independent [χ^2 (3, N=299)=28.68, p=.000]. As we can see in Table 1, men are less likely than women (as indicated by the negative sign on the parameter estimates) to have their marriage arranged, arranged-courtship, or courtship subject to parental approval compared to courtship type of marriage.

Furthermore, the marriage type was regressed on subsistence type to examine independence between the two. However, there are not enough cases in every subsistence category to allow any statistical test to be performed. To work around this problem, subsistence categories were collapsed. More specifically, agriculture, animal husbandry, and mixed agriculture-animal husbandry categories were collapsed into the agricultural–pastoral category. Additionally, hunting and gathering and mixed-hunting and gathering categories were collapsed into the predominantly-hunting and gathering category.

The marriage type for females was regressed on the collapsed subsistence type variable and the results indicate independence. Similarly, the marriage type for males was regressed on subsistence type, and the results indicate nonindependence at 10% level of significance [χ^2 (3, N=148)=6.70, p=.082]. In particular, the marriage of males in agricultural-pastoral societies in comparison to foraging ones is more likely to be of parental arrangement and parental arrangement-courtship than courtship type (Table 1).

We should note here that a multinomial regression model where both sex and subsistence type enter simultaneously in the regression equation, cannot be fitted for this data set. The reason being that for each society type in the sample there is information for the marriage of the females, the marriage of males and the subsistence type. Accordingly, in order for a model of the form

Table 1
Multinomial regression parameter estimates for marriage arrangements, males' marriage, decision makers, and marriage age variables

Explanatory variables								
Marriage arrangements	Parental arrangement		Courtship subject to parental approval		Courtship[a]		Parental arrangement-courtship	
	Parameter	Wald	Parameter	Wald	Parameter	Wald	Parameter	Wald
Males	−1.78	30.69***	−1.69	15.92***	–	–	−1.34	10.08**
Females[a]	–	–	–	–	–	–	–	–
Males' marriage	Parental arrangement		Courtship subject to parental approval		Courtship[a]		Parental arrangement-courtship	
	Parameter	Wald	Parameter	Wald	Parameter	Wald	Parameter	Wald
Agricultural-pastoral	1.0	5.29*	.506	1.16	–	–	.97	3.78*
Predominantly-hunting and gathering[a]	–	–	–	–	–	–	–	–
Decision makers	Males monopolise arrangements		Both participate males have more say		Both participate equal say[a]		Both participate females have more say	
	Parameter	Wald	Parameter	Wald	Parameter	Wald	Parameter	Wald
Agricultural-pastoral	1.40	3.79*	1.73	9.31**	–	–	.81	1.09
Predominantly-hunting and gathering[a]	–	–	–	–	–	–	–	–
Marriage age[b]	12–15		16–17		18–21		22–25	
	Parameter	Wald	Parameter	Wald	Parameter	Wald	Parameter	Wald
Males	–	–	1.16	2.51	2.02	8.70**	3.00	14.65***
Females[a]	–	–	–	–	–	–	–	–

[a] Indicates reference category.
[b] Note that very few observations were present in the sample for the younger and older ages. Thus to fit the model, the under 12 category was merged with 12–15 category and 22–25 category was merged with 26 and older category.
* Statistically significant at .05% level.
** Statistically significant at .01% level.
*** Statistically significant at .001% level.

marriage type=sex+subsistence to be fitted, we have to use subsistence data twice, once for the marriage of females and once for the marriage of males. This is inappropriate as we will end up having 372 observations for subsistence, while in reality, we only have 186 observations.

In Fig. 3 we can see that with respect to marriage arrangements, male parents and relatives have more decision making power than female ones across all society types. There is not a single case reported here in which females completely dominate arrangements. Furthermore, male parents and close kin appear to be more influential over marriage decisions in agricultural and pastoral societies than they are in hunting and gathering ones (Fig. 3).

To estimate whether males have more decision making power in nonforaging societies, multinomial regression was applied with the decision maker as a dependent variable and the subsistence type as an independent variable. Results indicate that the sex of the decision maker is contingent upon the subsistence type [χ^2 (3, N=83)=10.87, p=.012]. Males are more likely to monopolise and have more say than equal say in marriage arrangements in agricultural-pastoral societies (Table 1).

4.2. The offspring's mate choice

When marriages are arranged, the female is usually consulted by her parents but her consent is not always necessary for the marriage to proceed (Table 2). From the same table, it can be seen that it is less common for the marriage of males to proceed without the consideration of their opinion, while their consent to the marriage is more frequently necessary. However, this difference does not pass the significance level as multinomial regression indicates independence between the sex of the individual and the consent to marriage. Finally, the consent to marriage was regressed on subsistence type and the results indicate independence between the two.

Furthermore, across society types the most frequent female age at first marriage is between 12 and 15 years of age, whereas for males, it is between 16 and 21 (Table 2). Multinomial logistic regression was applied with the marriage age of the spouse as a dependent variable and the gender of the spouse as an independent variable. The results indicate nonindependence [χ^2 (3, N=87)=21.15, p=.000] with males being more likely than females to marry when they are 18–25 than when they are 12–15 (Table 1). Finally, the age at first marriage was regressed on subsistence type and the results indicate independence for both sexes.

In all society types, divorce can be obtained by both sexes (Table 2). In the cases where divorce cannot be obtained equally by both males and females, it is usually more difficult to be obtained by females. However, across all society types the most frequent case is that divorce is equally difficult to be obtained for both sexes. Multinomial regression of divorce difficulty on the subsistence type

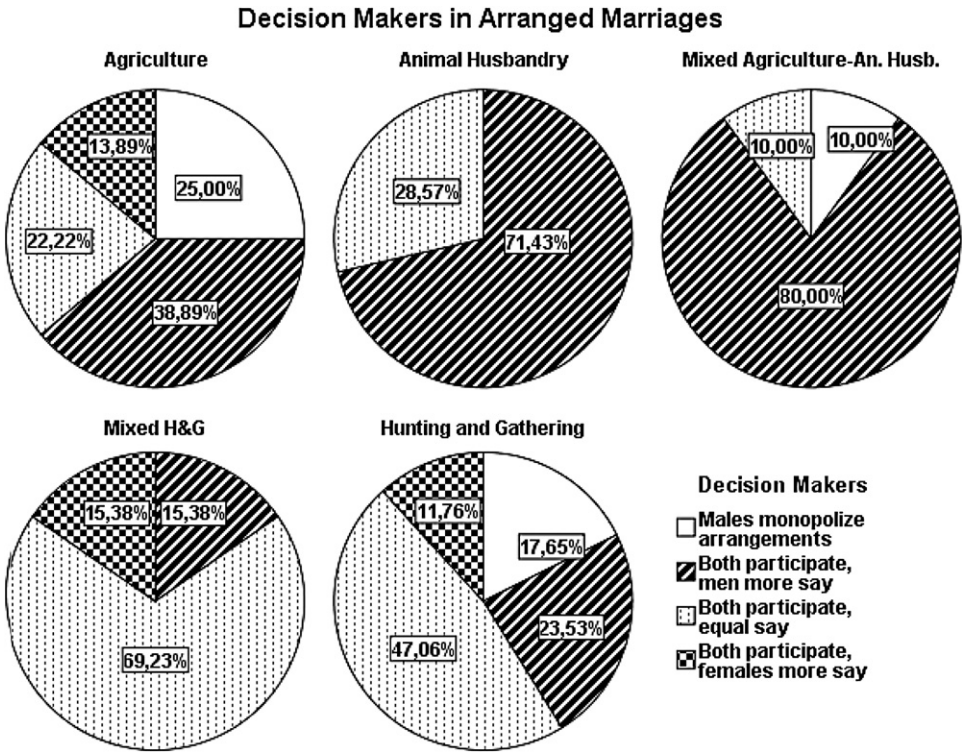


Fig. 3. The figures above depict decision makers in arranged marriages across different modes of subsistence. We can see that men dominate marriage arrangements in all society types.

Table 2
Divorce, marriage age at first marriages, and consent in arranged marriages

	Agriculture	Animal husbandry	Hunting and gathering	Mixed agriculture-animal husbandry	Mixed H&G
Divorce					
Males only	2 (5%)	–	1 (5.3%)	–	2 (10%)
Both more difficult for females	5 (12.5%)	1 (12.5%)	2 (10.5%)	2 (33.3%)	2 (10%)
Both	32 (80%)	6 (75%)	15 (78.9%)	3 (50%)	15 (80%)
Both more difficult for males	1 (2.5%)	1 (12.5%)	1 (5.3%)	1 (16.7%)	–
Divorce frequency					
Universal	4 (13.8%)	1 (14.3%)	1 (4.3%)	–	2 (10%)
Common	9 (31%)	2 (28.6%)	7 (30.4%)	3 (50%)	10 (50%)
Moderate	5 (17.2%)	1 (14.3%)	2 (8.7%)	1 (16.7%)	2 (10%)
Rare after the first years of marriage	2 (6.9%)	–	5 (21.7%)	–	3 (15%)
Rare-isolated instances-never	9 (31%)	3 (42.9%)	8 (34.8%)	2 (33.3%)	3 (15%)
Age at first marriage-females					
Under 12	–	–	1 (9.1%)	1 (20%)	–
12-15	8 (44.4%)	–	4 (36.4%)	2 (40%)	4 (44.4%)
16-17	2 (11.2%)	1 (50%)	4 (36.4%)	1 (20%)	3 (33.3%)
18-21	5 (27.8%)	1 (50%)	1 (9.1%)	1 (20%)	2 (22.2%)
22-25	3 (16.7%)	–	–	–	–
26-older	–	–	1 (9.1%)	–	–
Age at first marriage, males					
Under 12	–	–	–	–	–
12-15	3 (21.4%)	–	1 (9.1%)	1 (20%)	2 (20%)
16-17	5 (35.7%)	1 (50%)	2 (18.2%)	–	2 (20%)
18-21	3 (21.4%)	1 (50%)	5 (45.5%)	2 (40%)	2 (20%)
22-25	3 (21.4%)	–	–	2 (40%)	–
26-older	–	–	3 (27.3%)	–	4 (40%)
Consent Bride					
Necessary	9 (37.5%)	1 (33.3%)	4 (28.6%)	3 (50%)	2 (20%)
Consulted	8 (33.3%)	–	6 (42.9%)	1 (16.7%)	7 (70%)
Not consulted	7 (29.2%)	2 (66.7%)	4 (28.6%)	2 (33.3%)	1 (10%)
Consent groom					
Necessary	9 (39.1%)	–	8 (57.1%)	3 (60%)	3 (30%)
Consulted	9 (39.1%)	2 (100%)	4 (28.6%)	1 (20%)	6 (60%)
Not consulted	5 (21.7%)	–	2 (14.3%)	1 (20%)	1 (10%)
Female punishment					
No punishment	3 (14.3%)	1 (20%)	–	–	1 (7.7%)
Moderate	5 (23.8%)	2 (40%)	3 (25%)	1 (33.3%)	3 (23.1%)
Severe	13 (61.9%)	2 (40%)	9 (75%)	2 (66.7%)	9 (69.2%)

Note. The frequency is followed by the percentage in parenthesis. The percentages are estimated when only cases with nonmissing values are considered.

indicates independence between the two. Furthermore, divorce is usually common across all groups of societies (Table 2). Multinomial regression of divorce commonness on subsistence type also indicates independence.

Moreover, when the female is discovered to have committed adultery, she is usually punished by her husband (71% of the cases). Severe punishment, which may include the death of the female, is the most frequent form of punishment across societies, with no punishment or mild punishment being the rarest one (Table 2). Finally, multinomial regression indicates that female punishment is independent of subsistence type.

5. Discussion

The evidence presented here allows us to identify the typical patterns of mating in agricultural and pastoral societies. Consistent with the model of parental choice, parents exercise substantial control over the mating decisions of their offspring. Moreover, male parents have more decision making power over marriage arrangements than female parents, and the offspring exercise mate choice through various means, such as divorce and extramarital relationships. It is also found that parental control over mating is contingent upon the sex of the

offspring, and marriage is more frequently arranged for female than male offspring.

Additionally, matting patterns between foraging and agropastoral societies appear to be different in at least two ways: in agropastoral societies men have more decision making power over marriage arrangements than women, and parents exercise more control over the mating decisions of their male offspring. These differences can be explained by the wealth produced in each society type: more male-controlled wealth is produced in agropastoral societies than in hunting and gathering ones, which in turn provides men with more power over their offspring's mating decisions.

Given that, in a majority of societies, wealth is inherited through the male line from father to son (Hartung, 1976, 1982), a man risks forfeiting his inheritance rights if he does not comply with his father's wishes. This gives fathers extra means of controlling their sons' mate choices. Also, resources inherited in agropastoral societies are likely to be substantial in comparison to resources inherited in foraging ones. This gives an extra motive to a son to comply with his father's wishes.

The data presented here provide evidence that the context of human mating over the last 10,000 years should and could have influenced sexual selection in the way predicted by the model of parental choice. In particular, as male parents have become more dominant in controlling their children's mating decisions, evolutionary pressure on mate seekers to appeal to male parents should have increased during the Holocene. In turn, traits which made an individual more likely to be selected as a spouse for their offspring by male parents should have also increased in the population under strong directional selection. Future research should test these hypotheses by providing direct evidence of parental preference and traits in potential mates that have evolved specifically in response to these preferences.

This research is not without limitations. The use of pre-existing codes that were not constructed with the parental choice theory in mind is advantageous. However, there are also certain limitations. For example, the coding of marriage refers to the offspring's first marriage. In some societies subsequent marriages are also arranged by parents, while in others, parental control substantially diminishes after the first marriage. Thus, although first marriage is a good proxy for assessing parental control, if the full extent of parental choice is to be estimated, coding for subsequent marriages is also desirable.

In addition, data for many societies are not available for a number of variables. This may produce a certain bias: for instance these variables may appear to be independent of the type of society while in fact they are not. Finally, it is assumed here that the labels used by ethnologists to categorize marriage systems constitute accurate descriptions of the distribution of marriage types within a given society. This does not exclude the possibility that the ethnologists refer to what is considered formal or most appropriate in a given society rather than what actually happens.

To conclude, the anthropological record indicates that sexual selection in our species has certain unique features. Evidently, humans are the only species on the planet where males are selected by other males for reproductive purposes. This pattern of mating is typical not only of hunting and gathering societies, but also of agricultural and pastoral ones. Ultimately, there are good reasons to believe that parental choice was (and still is) a significant sexual selection force during most of human evolution. Future research should therefore attempt to provide evidence that certain traits have been sexually selected in the way predicted by the model of parental choice.

Acknowledgments

I am in debt to Lia Mexa, Stylianos Papanastasiou, Aris Boukouras, Elpiniki Fragouli, and Evangelos Aspiotis for their help and advice during the preparation of this manuscript. Finally, I would like to thank Daniel Fessler and two anonymous reviewers for their constructive criticisms and feedback which enabled me to improve the quality of my work.

References

Apostolou, M. (2007a). Elements of parental choice: The evolution of parental preferences in relation to in-law selection. *Evolutionary Psychology*, *5*, 70–83.

Apostolou, M. (2007b). Sexual selection under parental choice: The role of parents in the evolution of human mating. *Evolution and Human Behavior*, *28*, 403–409.

Apostolou, M. (2008a). Bridewealth and brideservice as instruments of parental choice. *Journal of Social, Evolutionary, and Cultural Psychology*, *2*, 89–102.

Apostolou, M. (2008b). Parent-offspring conflict over mating: The case of beauty. *Evolutionary Psychology*, *6*, 303–315.

Apostolou, M. (2008c). Parent-offspring conflict over mating: The case of family background. *Evolutionary Psychology*, *6*, 456–468.

Armelagos, G. J., & Harper, K. N. (2005). Genomics at the origin of agriculture, part one. *Evolutionary Anthropology*, *14*, 68–77.

Broude, G. J., & Green, S. J. (1983). Cross-cultural codes on husband-wife relationships. *Ethnology*, *22*, 263–280.

Buunk, A. P., Park, J. H., & Dubbs, S. L. (2008). Parent-offspring conflict in mate preferences. *Review of General Psychology*, *12*, 47–62.

Ember, C. R. (1978). Myths about hunter–gatherers. *Ethnology*, *17*, 439–448.

Endicott, K. (1999). Gender relations in hunter–gatherer societies. In R. B. Lee, & R. Daly, (Eds.), *The Cambridge encyclopedia of hunters and gatherers* (pp. 411–418). Cambridge: Cambridge University Press.

Fisher, R. A. (1958). *The genetic theory of natural selection*. 2nd ed. Oxford: Oxford University Press.

Flinn, M. V., & Low, B. S. (1986). Resource distribution, social competition and mating patterns in human societies. In D. I. Rubenstein, & R. W. Wrangham, (Eds.), *Ecological aspects of social evolution* (pp. 217–243). Princeton: Princeton University Press.

Hartung, J. (1976). On natural selection and the inheritance of wealth. *Current Anthropology*, *17*, 607–613.

Hartung, J. (1982). Polygyny and inheritance of wealth. *Current Anthropology*, *23*, 1–12.

Hawks, J., Wang, E. T., Cochran, G. M., Harpending, H. C., & Moyzis, R. K. (2007). Recent acceleration of human adaptive evolution. *Proceedings of the National Academy of Sciences*, *104*, 20753–20758.

Lee, R. B., & DeVore, I. (1968). *Man the hunter*. New York: Aldine.

Lerner, G. (1986). *The creation of patriarchy*. Oxford: Oxford University Press.

Minturn, L., Grosse, M., & Haider, S. (1969). Cultural patterning of sexual beliefs and behaviour. *Ethnology*, *8*, 301–318.

Murdock, G. P. (1981). *Atlas of world cultures*. Pittsburgh: University of Pittsburgh Press.

Perilloux, C., Fleischman, D. S., & Buss, D. M. (2008). The daughter guardinghypothesis: Parental influence on, and emotional reaction to, offspring's mating behavior. *Evolutionary Psychology*, *6*, 217–233.

Price, T. D. (2000). *Europe's first farmers*. Cambridge: Cambridge University Press.

Smuts, B. (1995). The evolutionary origins of patriarchy. *Human Nature*, *6*, 1–32.

Stephens, W. N. (1963). *The family in cross-cultural perspective*. New York: Holt, Rinehart & Winston.

Trivers, R. (1974). Parent offspring conflict. *American Zoologist*, *24*, 249–264.

Trivers, R. L. (1972). Parental investment and sexual selection. In B. Campbell, (Ed.), *Sexual selection and the descent of man: 1871-1971* (pp. 136–179). Chicago: Aldine.

Westermarck, E. (1925). *The history of human marriage*. London: Macmillan.

Whyte, M. K. (1978). *The status of women in preindustrial societies*. Princeton, NJ: Princeton University Press.

World-Cultures CD-Rom. (2006). *Journal of Comparative Cross Cultural Research*.